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-- MYSQL DATA TYPES LAB

-- Covers: Character, Binary, Text, Blob, Numeric,

--     Date/Time, ENUM, SET

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-- =====

-- 1. CREATE DATABASE

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DROP DATABASE IF EXISTS datatype_lab;

CREATE DATABASE datatype_lab;

USE datatype_lab;

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-- 2. CHARACTER & BINARY DATA TYPES

-- CHAR, VARCHAR, BINARY, VARBINARY

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CREATE TABLE char_types (

    id INT AUTO_INCREMENT PRIMARY KEY,

    -- Fixed-length string (always uses full size)

    fixed_char CHAR(10),

    -- Variable-length string (saves space)

    variable_char VARCHAR(100),

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-- Fixed-length binary data
fixed_binary BINARY(5),

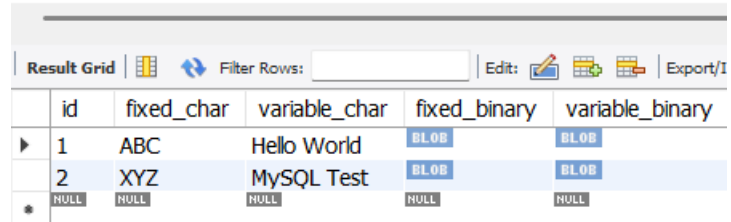
-- Variable-length binary data
variable_binary VARBINARY(50)
);

-- Insert sample data
INSERT INTO char_types (fixed_char, variable_char, fixed_binary, variable_binary)
VALUES
('ABC', 'Hello World', '12345', 'BinaryData1'),
('XYZ', 'MySQL Test', '54321', 'BinaryData2');

```

-- View data

41 • **SELECT * FROM char_types;**



	id	fixed_char	variable_char	fixed_binary	variable_binary
▶	1	ABC	Hello World	BLOB	BLOB
	2	XYZ	MySQL Test	BLOB	BLOB
*		NULL	NULL	NULL	NULL

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-- 3. TEXT & BLOB DATA TYPES

-- TINYTEXT, TEXT, MEDIUMTEXT, LONGTEXT

-- TINYBLOB, BLOB, MEDIUMBLOB, LONGBLOB

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CREATE TABLE text_blob_types (
    id INT AUTO_INCREMENT PRIMARY KEY,

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-- Text types (store large text)

tiny_text TINYTEXT,

text_data TEXT,

medium_text MEDIUMTEXT,

long_text LONGTEXT,


-- Binary large objects (images, files, etc.)

tiny_blob TINYBLOB,

blob_data BLOB,

medium_blob MEDIUMBLOB,

long_blob LONGBLOB

);


-- Insert sample data

INSERT INTO text_blob_types

(tiny_text, text_data, medium_text, long_text, tiny_blob, blob_data)

VALUES

('Short text', 'This is normal text', 'Medium text example', 'Long text example',

'abc', 'binarydata');

```

```

-- View selected columns

```

72 • **SELECT** id, tiny_text, text_data **FROM** text_blob_types;

Result Grid | Filter Rows: | Edit: | Export/Import: | Wrap

	id	tiny_text	text_data
▶	1	Short text	This is normal text
*	NULL	NULL	NULL

```

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-- 4. NUMERIC DATA TYPES

-- BIT, TINYINT, BOOLEAN, SMALLINT, MEDIUMINT, INT, BIGINT

-- FLOAT, DOUBLE, DECIMAL

-- =====

CREATE TABLE numeric_types (

    id INT AUTO_INCREMENT PRIMARY KEY,


    -- Bit value (binary representation)

    bit_val BIT(4),


    -- Integer types

    tiny_val TINYINT,

    bool_val BOOLEAN,

    small_val SMALLINT,

    medium_val MEDIUMINT,

    int_val INT,

    big_val BIGINT,


    -- Floating point (approximate values)

    float_val FLOAT(10,2),

    double_val DOUBLE(10,2),


    -- Exact numeric (financial data)

    decimal_val DECIMAL(10,2)

);

```

-- Insert sample data

INSERT INTO numeric_types

(bit_val, tiny_val, bool_val, small_val, medium_val, int_val, big_val, float_val, double_val, decimal_val)

VALUES

(b'1010', 10, TRUE, 100, 1000, 10000, 1000000, 10.55, 20.12, 999.99),

(b'0101', 5, FALSE, 200, 2000, 20000, 2000000, 15.75, 30.56, 888.88);

-- View data

109 • **SELECT** * **FROM** numeric_types;

Result Grid Filter Rows: Edit: Export/Import: Wrap Cell Content:											
	id	bit_val	tiny_val	bool_val	small_val	medium_val	int_val	big_val	float_val	double_val	decimal_val
▶	1	10	10	1	100	1000	10000	1000000	10.55	20.12	999.99
	2	5	5	0	200	2000	20000	2000000	15.75	30.56	888.88
✱	NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL

-- =====

-- 5. DATE & TIME DATA TYPES

-- DATE, DATETIME, TIMESTAMP, TIME, YEAR

-- =====

CREATE TABLE datetime_types (

id INT AUTO_INCREMENT PRIMARY KEY,

-- Date only

date_val DATE,

-- Date and time

datetime_val DATETIME,

```

-- Auto timestamp (updated automatically)
timestamp_val TIMESTAMP DEFAULT CURRENT_TIMESTAMP,

-- Time only
time_val TIME,

-- Year
year_val YEAR
);

-- Insert sample data
INSERT INTO datetime_types (date_val, datetime_val, time_val, year_val)
VALUES
('2025-01-01', '2025-01-01 10:30:00', '10:30:00', 2025),
('2026-06-15', '2026-06-15 15:45:00', '15:45:00', 2026);

-- View data

```

141 • **SELECT** * **FROM** datetime_types;

id	date_val	datetime_val	timestamp_val	time_val	year_val
1	2025-01-01	2025-01-01 10:30:00	2026-05-11 13:46:53	10:30:00	2025
2	2026-06-15	2026-06-15 15:45:00	2026-05-11 13:46:53	15:45:00	2026
*	NULL	NULL	NULL	NULL	NULL

```
-- =====  
  
-- 6. ENUM & SET DATA TYPES  
  
-- ENUM = single choice  
  
-- SET = multiple choices  
  
-- =====  
  
CREATE TABLE enum_set_types (  
    id INT AUTO_INCREMENT PRIMARY KEY,  
  
    -- ENUM allows only one value  
    status ENUM('Pending', 'Approved', 'Rejected'),  
  
    -- SET allows multiple values  
    permissions SET('Read', 'Write', 'Execute')  
);  
  
-- Insert sample data  
  
INSERT INTO enum_set_types (status, permissions)  
VALUES  
('Pending', 'Read'),  
('Approved', 'Read,Write'),  
('Rejected', 'Read,Execute');
```

166 • **SELECT * FROM** enum_set_types;

Result Grid   Filter Rows: Edit:  

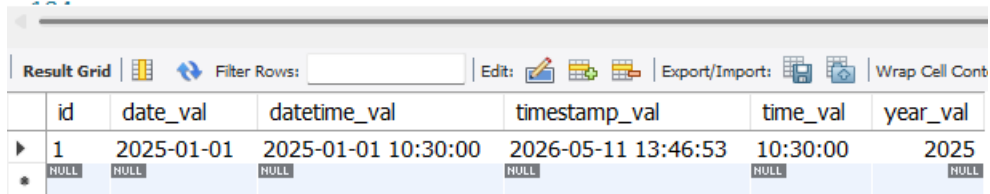
	id	status	permissions
▶	1	Pending	Read
	2	Approved	Read,Write
	3	Rejected	Read,Execute
*	NULL	NULL	NULL

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[illegible][illegible]

-- Date filter

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182 • SELECT * FROM datetime_types
183 WHERE year_val = 2025;
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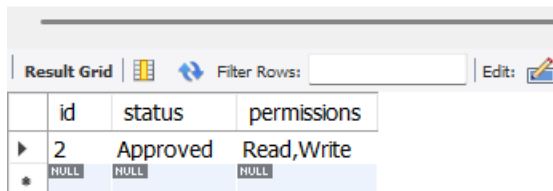


The screenshot shows a database query result grid. The grid has a header row with columns: id, date_val, datetime_val, timestamp_val, time_val, and year_val. The first data row shows id=1, date_val=2025-01-01, datetime_val=2025-01-01 10:30:00, timestamp_val=2026-05-11 13:46:53, time_val=10:30:00, and year_val=2025. Below the first data row is a row with asterisks and NULL values for each column.

	id	date_val	datetime_val	timestamp_val	time_val	year_val
▶	1	2025-01-01	2025-01-01 10:30:00	2026-05-11 13:46:53	10:30:00	2025
*	NULL	NULL	NULL	NULL	NULL	NULL

-- ENUM filter

```
186 • SELECT * FROM enum_set_types
187 WHERE status = 'Approved';
```



The screenshot shows a database query result grid. The grid has a header row with columns: id, status, and permissions. The first data row shows id=2, status=Approved, and permissions=Read,Write. Below the first data row is a row with asterisks and NULL values for each column.

	id	status	permissions
▶	2	Approved	Read,Write
*	NULL	NULL	NULL

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-- 8. CLEANUP (OPTIONAL)

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-- DROP DATABASE datatype_lab;